ENERGY MARKETS & INFRASTRUCTURE

Gas Market Design & Architecture

Liberalization, Grid Infrastructure, Operational Conventions, and European Supply & Demand Dynamics

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Unbundled Grid Operators

Wholesale Trading Mechanics

EU Single Price Vision

MARKET EVOLUTION

The Path to Gas Market Liberalization

Monopoly to Open Competition

ERA 1 • BILATERAL

Oil-Indexed Monopolies

Gas Sale Purchase Agreements (GSPA) indexed on oil exchanged strictly between national monopoly pairs.

1979 • UK REFORMS

Unbundling Initiative

Separated Transport, Distribution, Storage, and Imports from competitive market sales to stop market abuse.

2000 • EU TRANSITION

Open EU Gas Market

Mandated third-party access and cross-border interconnections across European Union member states.

END GOAL • INTEGRATION

Single European Price

Frictionless cross-border flow creating liquid virtual trading points with a unified wholesale price signal.

SYSTEM ARCHITECTURE

Grid Infrastructure & Network Operators

TSO Transmission

High-pressure, cross-border interconnectors. Functions as the virtual trading hub establishing wholesale prices.

DSO Distribution

Low-pressure local pipeline networks delivering gas directly to industrial and residential end-users.

SSO Storage System

Underground gas storage facilities providing critical seasonal balancing between summer and winter.

LSO LNG Terminals

Liquefied Natural Gas import, regasification, and entry facilities feeding national networks.



TARIFF MECHANISM

Using each network as well as entering and exiting one is subject to regulated entry/exit fees paid to system operators.

COMMERCIAL MECHANICS

Traded Products, Prices & Market Conventions

Operational Cycles

Gas Day Definition

Standard daily trading interval runs from **05:00 AM to 05:00 AM** next day (Day Ahead).

Power vs. Energy Units

Gas volume traded in energy units (MWh or therms) distinguishing rate vs total volume.

Gas & Storage Years

October to September cycle driving seasonal injection and withdrawal spreads.

Financial & Forward Curves

Futures Contracts

Fixed-for-floating derivative swaps locking in forward price risk across curve tenors.

Seasonality Spreads

Winter premium over Summer pricing reflecting heating demand surges.

Hub Benchmarks

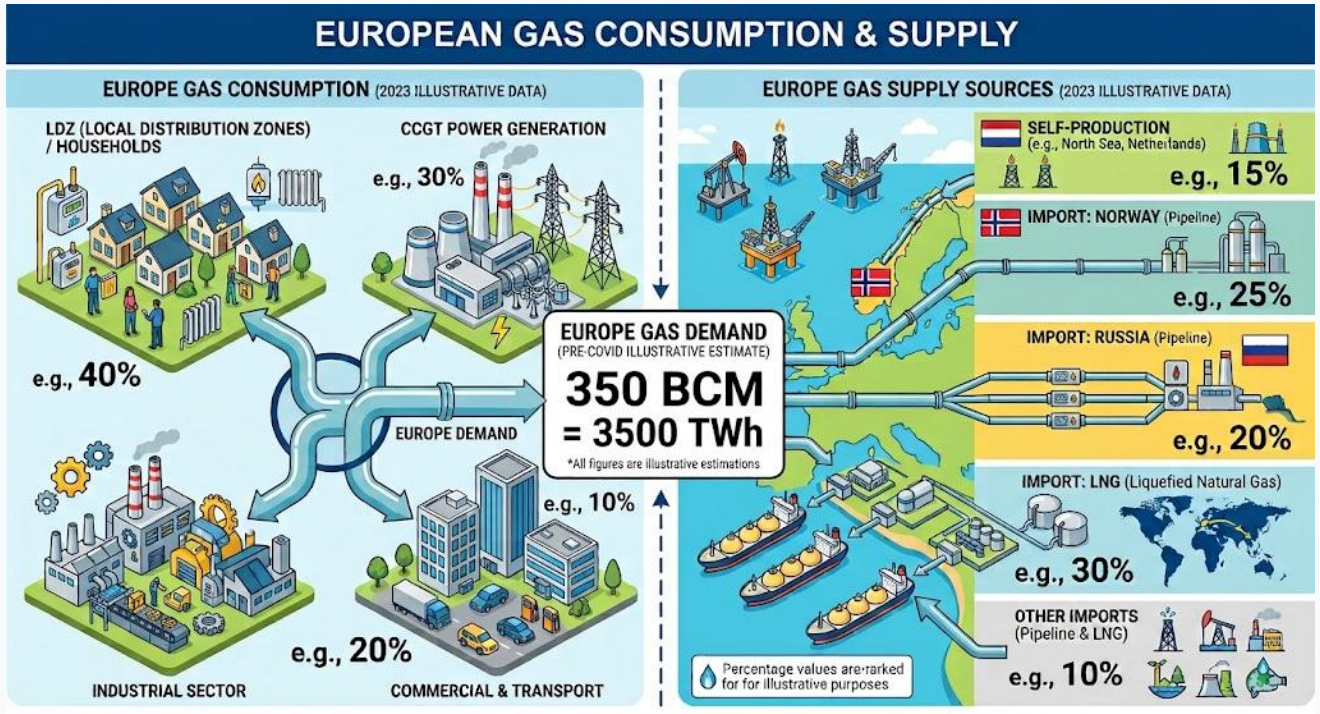
Dutch TTF and regional VTPs providing transparent forward market curve signals.



FUNDAMENTALS

European Gas Consumption & Supply Balance

Annual Demand: ~350 BCM = 3,500 TWh



Key Demand Sectors

40% Households (LDZ)	**30%** CCGT Power Gen
20% Industrial Sector	**10%** Commercial / Transport

Major Supply Mix

LNG Imports	~30%
Norway Pipelines	~25%
Russia Pipelines	~20%
Self-Production (North Sea/NL)	~15%